1 Quick Review

Switching gears from eigenproblems and diagonalization, we've already seen that linearly independent vectors are a very important topic in linear algebra. However, quantifying linea independence has ben a topic we've avoided thus far. Is there a way we can mathematically represent two vectors being very different or very similar?

That's where the **inner product** comes in. You have been introduced to an inner product before under the name of the *dot product*:

$$\langle u, v\rangle = u \cdot v = u_1v_1 + u_2v_2 + \cdots + u_nv_n$$

Like vector spaces, the inner product is defined with axioms. One may define an inner product in any way they wish, as long as it follows the axioms.

Definition 1: Inner Product

$\forall\, k \in \mathbb{R}$ and vectors u, v, w in a vector space V, the inner product associates a scalar value with a pair of any vectors in V. To be an inner product, the operation must follow the following axioms:

1. $\langle u, v\rangle = \langle v, u\rangle$
2. $\langle u + v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$
3. $\langle ku, v\rangle = k\langle u, v\rangle$
4. $\langle u, u\rangle \geq 0$ and $\langle u, w\rangle = 0$ iff $u = 0$

A vector space with a defined inner product is called an inner product space.

By using the inner product we can define the **norm** (or more informally, the *length*) of a vector. This is often denoted as $||v||$ or sometimes just $|v|$, where v is a vector. The most general definition is:

$$||v|| = \sqrt{\langle v, v\rangle}$$

Another important application of the inner product is the notion of angle or orthogonality. This is where we can define a relationship between two vectors. The angle between two vectors is derived from the **Cauchy-Schwarz Inequality**:

$$|\langle u, v\rangle| \leq ||u||\,||v||.$$

As a result we can say that the cosine of the angle between two vectors is:

$$\cos(\theta) = \frac{\langle u, v\rangle}{||u||\,||v||}$$

A relatively simple, but very helpful conclusion is that two vectors are orthogonal if the inner product between them is equal to 0. We can apply this conclusion to much more abstract vector spaces and calculate the orthogonality algebraically.

2 Problems

1. Let $v_1 = (4+3i,\ 1-i)$ and $v_2 = (-5-3i,\ i)$ where $v_1, v_2 \in \mathbb{C}^2$.

   (a) Using the standard inner product on $\mathbb{C}^2$, evaluate $\langle v_1, v_2 \rangle$.

   (b) Let $\mathbf{A} = [\, v_1 \mid v_2 \,]$ be the 2×2 matrix formed by v_1 and v_2. Find $\mathrm{Re}(\mathbf{A})$.

   (c) Find the eigenvalues and eigenvectors of $\mathrm{Re}(\mathbf{A})$.

2. **(PROOF).** Let $u = u_1 + u_2x + u_3x^2$ and $v = v_1 + v_2x + v_3x^2$, where $u, v \in \mathbb{P}_2$. If $\langle u, v \rangle := 3u_1v_1 + 5u_2v_2 + u_3v_3$, prove whether or not $\langle u, v \rangle$ is a valid inner product.

3. Let $f = f(x)$ and $g = g(x)$ be two functions in $\mathbb{C}[-a, a]$.

   (a) Let f be an *odd* function and g be an *even* function. Recalling what happens when you integrate an even or odd function, what can be said about the orthogonality of $f(x)$ and $g(x)$? What about their linear independence?

   (b) Let $f(x) = x^2$ and $g(x) = x$. Find the cosine of the angle between $f(x)$ and $g(x)$.

4. Let there be a set $S = \{(2,0,1),(1,4,0),(4,8,1),(-5,-4,-2)\} \in \mathbb{R}^3$. Find the orthogonal complement $S^\perp$ of the set S.

5. *All of the following information was collected with the help of Dr. Trefor Bazett's* *The beautiful geometric view of FOURIER SERIES // The Linear Algebra Perspective*. Due to the the speeds at which electricity transmits, electrical engineers often deal with discontinuous, periodic functions such as the *square wave*. Period means that the function repeats or more formally, $f(t+T) = f(t)$.

   However, dealing with the mathematics of discontinuous functions is often more troublesome than the mathematics of continuous functions. For this reason, mathematicians often use the *Fourier Series* to approximate discontinuous, periodic functions with trigonometric functions:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi t}{L}\right) + \sum_{n=1}^{\infty} b_n \cos\left(\frac{n\pi t}{L}\right)$$

   (a) Let's isolate the middle term. The derivation isn't important for our purposes, but it can be found that

$$a_m = \frac{1}{L}\int_{-L}^{L} f(t)\cos(mt)dt.$$

   This formula looks very similar to that of the integral inner product $\langle f, g\rangle$ on $\mathbb{C}[a,b]$. Identify f, g, a, and b.

   $f =$ ______ $g =$ ______ $a =$ ______ $b =$ ______

   (b) If $f(t)$ and $\cos(mt)$ were *orthogonal*, what would a_m be for all m? What can be said about the parity of $f(t)$?

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   (c) Using what you know about trigonometric functions, is the equation for $f(t)$ a basis for periodic functions? In other words, can the terms take on any period and are sine and cosine linearly independent of one another? If it could be a basis, what would be its dimension?

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